

CIB Mango Tree

DETECTING COORDINATED BEHAVIOR ON SOCIAL MEDIA

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The George Washington University

Capstone Project - Fall 2025





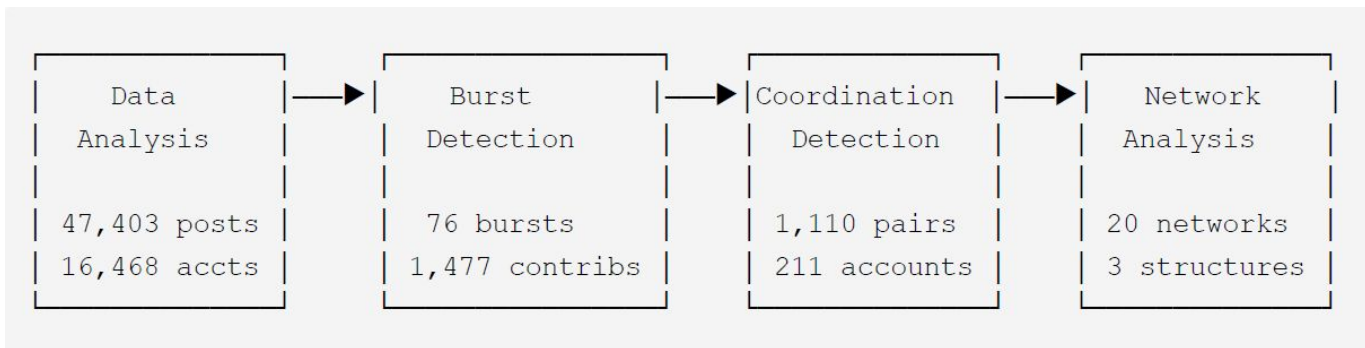
The “Black Box” Problem

- The Threat: Coordinated Inauthentic Behavior (CIB) is scaling rapidly
- The Blind Spot: We lack “Ground Truth” labels. We don’t know who the bad actors are
- The Question: Without labels, how do we scientifically distinguish a **coordinated attack** from organic trend?



The Methodology

A 4-stage pipeline built on 47,403 Truth Social posts

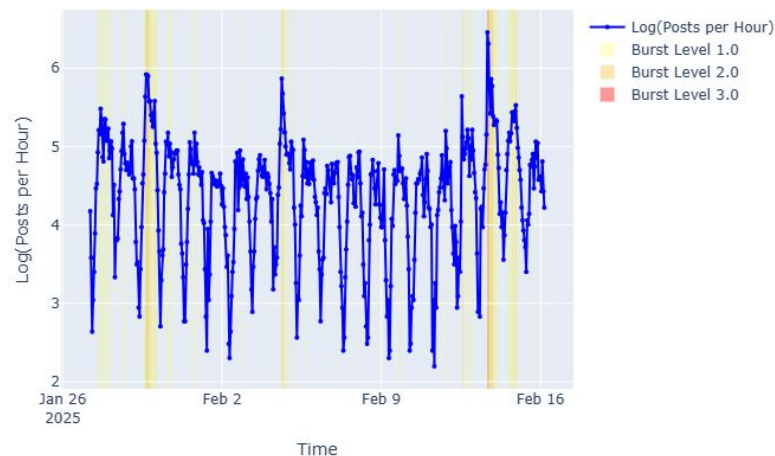




Burst Detection

- Kleinberg's Algorithm: Identifies time windows where activity spikes above baseline
- Core Idea: Contributor Selection
- The Result: we filtered out casual users, reducing noise from thousands of users to just active few hundred participants in bursts.

Burst Detection: Posts per Hour (Log Transformed) with Burst Regions





Coordination Analysis

I tested 5 signals in a “Phase Additive” framework:

Phase 1-4 (Accepted):

- ✓ Content Similarity
- ✓ Hashtag Coordination
- ✓ URL Sharing
- ✓ Retweet Amplification
- ✓ Temporal Synchronization

Phase 5 (Rejected):

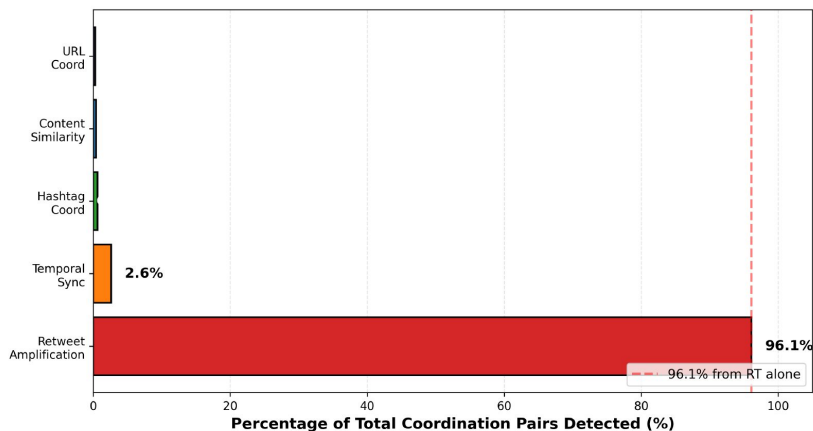
Signal: “Mechanical Posting”

Result: Double detection (+105%) but introduced **False Positives** due to timezone, behavioral coincident.



Key Findings: Signal Hierarchy

Signal Contribution Hierarchy:
Retweet Amplification Dominates Coordination Detection

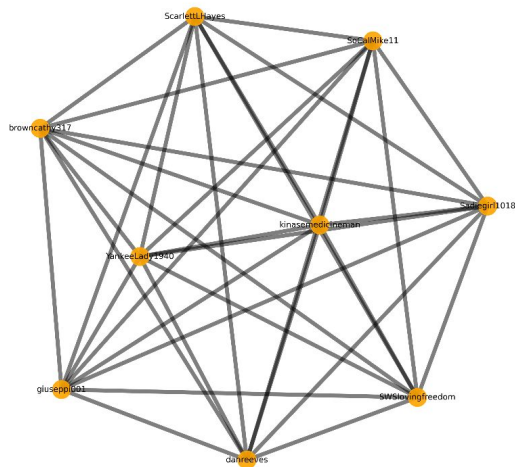


Retweets (96.1%): The “smoking gun.” The primary mechanism of coordination.

Content signals (~1.3%): Surprisingly low impact. Coordinated actors rarely copy and paste text exactly



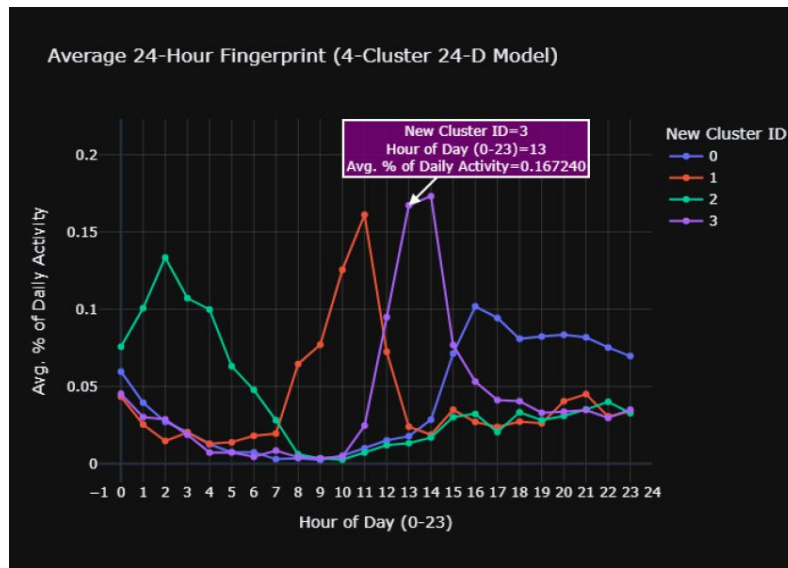
From pairs to Networks



- **Input:** 1,110 Coordination Pairs detected.
- **The Logic:** Pairs become Weighted Edges in NetworkX
- **Edge Weight:** Maximum Confidence score
- **Result:** Collapsed pairs into 20 distinct Connected Networks

Alternative: Temporal Clustering

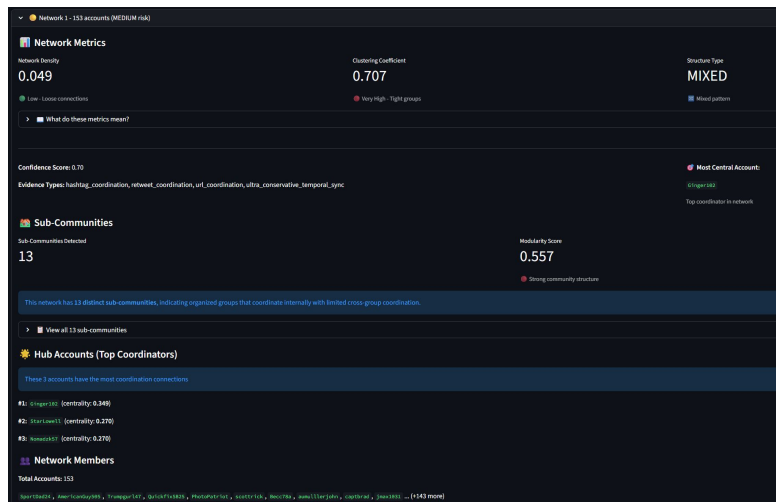
- Do bots share “Temporal Personas”?
- **Method:** K-Means Clustering
- **Result:** found 4 behaviors
- **Limitation:** Without original timezone metadata (UTC issue), we cannot verify if this is coordination





Network Analysis

- We detected 20 networks
- Network 1 is (153 accounts) is dominant
- Conclusion: organized sub-communities, not a random botnet





The Dashboard



Conclusion

- **Retweets are King:** Retweet amplification accounts for 96.1% of detections. Complex content analysis adds minimal value.
- **Rigor over Quantity:** We rejected behavioral patterns (Phase 5) because precision matters more than high detection counts.
- Coordination is not random